

Shraddha Harihar

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EDUCATION

- **M.Sc. Mathematics** Aug 2023 - May 2025
Dr. Vishwanath Karad MIT World Peace University, Pune, India GPA: 9.15/10
- **B.Sc. (Hons) Mathematics** Aug 2020 - May 2023
M. S. Ramaiah University of Applied Sciences, Bangalore, India GPA: 8.78/10

RESEARCH EXPERIENCE

- **k -Domination Vertex Connectivity of Connected Graphs** Aug 2023 - Dec 2024
MIT World Peace University *Pune, India*
 - **Concept Introduction:** Introduced and defined the novel parameter k -domination vertex connectivity, integrating graph domination and vertex connectivity.
 - **Mathematical Results:** Established theorems, exact formulas, and bounds for paths, cycles, and regular graphs using rigorous proof techniques.
 - **Computational Verification:** Developed Python code to verify results and explore concrete examples.
 - **Paper Finalization:** Finalized a manuscript for submission to a peer-reviewed mathematical journal.
- **Applications of Number Theory in Cryptography** May 2023
M. S. Ramaiah University of Applied Sciences *Bangalore, India*
 - **Cryptographic Foundations:** Analyzed the foundational role of number theory (prime factorization, modular arithmetic, Discrete Logarithm Problem) in modern public-key cryptosystems like RSA and Diffie-Hellman.
 - **Cryptographic Security:** Investigated how security relies on the computational hardness of problems like integer factorization and the DLP.
 - **Algorithmic Foundations:** Explored applications of Euler's theorem, Miller-Rabin primality testing, and properties of primitive roots in cryptographic protocols and security proofs.
 - **MATLAB Implementation:** Implemented RSA encryption in MATLAB to demonstrate practical use of theoretical principles.

CONFERENCES

- **Oral Presentation: RAPIAM-2025** 5-6 June, 2025
JSPM University in collaboration with the Indian Mathematical Society. *Pune, India*
 - **Paper Title:** On the k -Domination Vertex Connectivity of Connected Graphs.

SKILLS

- **Mathematics:** Graph Theory, Probability and Statistics, Linear Algebra, Number Theory, Cryptography
- **Machine Learning:** Supervised and Unsupervised learning principles, Neural Networks, Mathematical ML Theory, Data Preprocessing
- **Programming:** Python, R, MATLAB, SQL
- **Libraries and Frameworks:** NumPy, SciPy, Pandas, Matplotlib, TensorFlow, Keras, OpenCV, NLTK, Scikit-learn, Seaborn
- **Tools:** LaTeX, Jupyter Notebook, Spyder, RStudio, VS Code, Git/Github, SageMath

PROJECTS

- **Morse code detection using Eye Blinks (June 2024):** Created a real-time Morse code translator via eye blinks (Python) using computer vision libraries (MediaPipe, OpenFace) for facial landmark tracking and blink detection.
- **Tic-Tac-Toe (vs AI): Minimax Algorithm (June 2024):** Developed a graphical Tic-Tac-Toe game (Python, Pygame) featuring an AI opponent implementing the Minimax algorithm with game state management for optimal move selection.
- **Sentiment Analysis on Large-Scale Text Data (April 2024):** Applied NLP techniques (NLTK, TextBlob, VaderSentiment) in Python to analyze sentiment on a large dataset, performing text preprocessing, scoring, and visualizing key patterns (e.g., word clouds).

WORK EXPERIENCE

- **Research Analyst** Feb 2025 - Feb 2026
MyAnatomy Integration
 - **AI/ML Integration Research:** Investigated AI/ML applications for recruitment platforms, evaluating LLMs, chatbots, and NLP APIs (including Gemini) for potential integration into product workflows.
 - **Applied NLP Prototyping:** Developed proof-of-concept NLP prototypes such as job-role extraction and automated question-generation systems to assess feasibility and model performance.
 - **Platform Data Analysis:** Analyzed platform data, user interactions to identify technical bottlenecks and opportunities for AI-driven improvements.
 - **Applied AI Enhancements:** Proposed ML-based enhancements that informed platform strategy and strengthened applied research skills in practical AI deployment.
- **Data Science Intern** April 2024
Prodigy Infotech
 - **Exploratory Data Analysis:** Engaged with customer demographic and behavioral data, applying EDA techniques, including cleaning and visualization, to uncover patterns and subsequently build and evaluate a decision tree classifier.
 - **Sentiment Analysis:** Used NLP tools (NLTK, TextBlob) to analyze sentiment in customer food reviews, gaining experience in transforming unstructured text into meaningful insights.

ACHIEVEMENTS

- **Best Research Project Award:** Recognized for outstanding research contributions in the M.Sc. Mathematics program for work on k -domination vertex connectivity of connected graphs (MIT-WPU, May 2025).
- **Winner:** Ideathon, Hackathon (MIT-WPU, March 2024). Focused on object detection using YOLO algorithm.
- **Academic Excellence Awards, MIT-WPU:**
 - **The Dr. Vishwanath Karad scholarship (100% Merit):** Awarded to the top 1% students for academic excellence, 2023-2025.
 - **Silver Medal:** Awarded for attaining a top-tier GPA in M.Sc. Mathematics cohort, May 2025.